

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

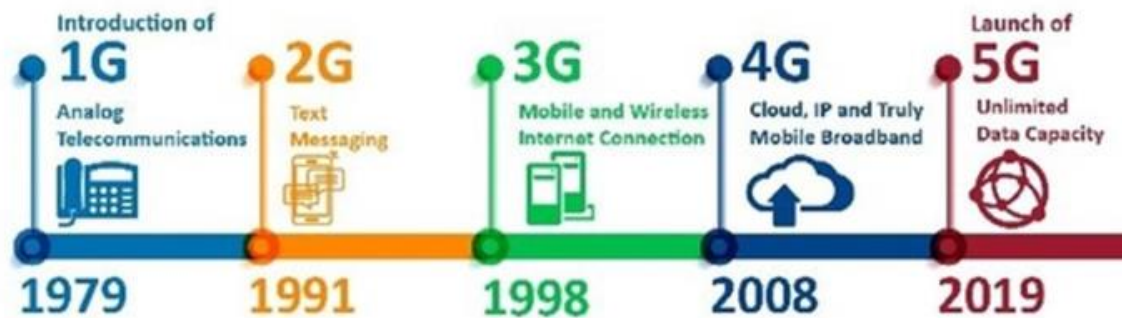
MODULE - 31

INFORMATION & COMMUNICATION TECHNOLOGY (ICT) - 12

5G TECHNOLOGY

- 5G is the 5th generation mobile network. It is a new global wireless standard after 1G, 2G, 3G, and 4G networks. 5G enables a new kind of network that is designed to connect virtually everyone and everything together including machines, objects, and devices (IOT).
- 5G technology is meant to deliver higher multi-Gbps peak data speeds, ultra-low latency, more reliability, massive network capacity, increased availability, and a more uniform user experience to more users.
- 5G will provide download speed of 1 Gbps, which is at least 100 times the existing data speeds.
- 5G will have a wider area in the frequency spectrum (range of frequencies) that will ensure no network congestion.

The Evolution of 5G



INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

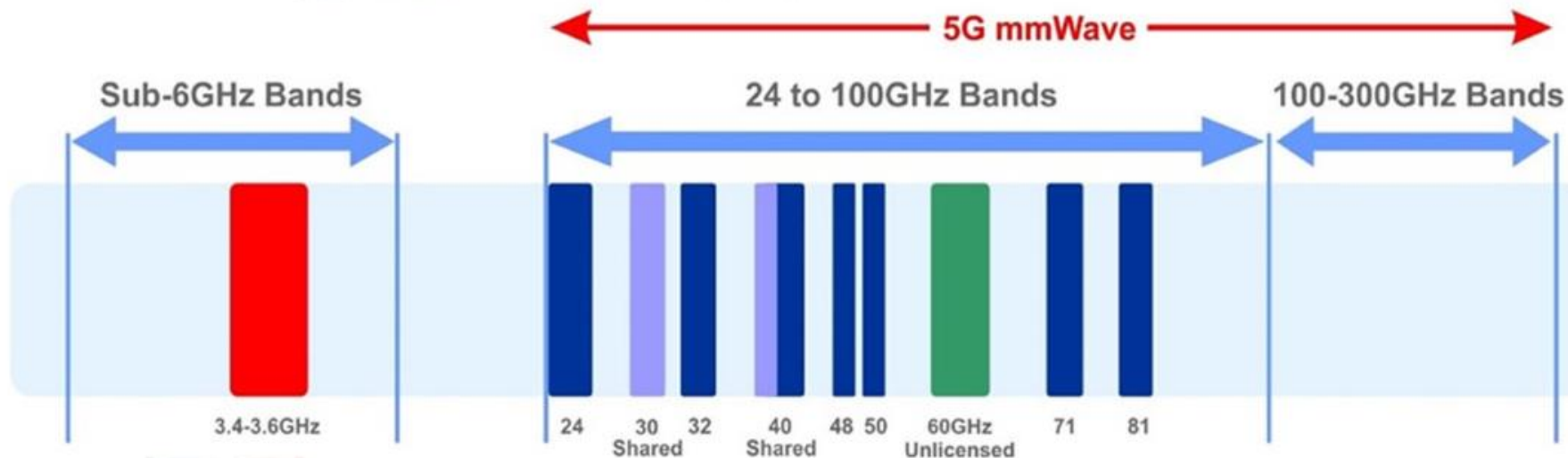
5G TECHNOLOGY

- The standards for the usage of 5G are defined and driven by 3rd Generation Partnership Project.
- The 3rd Generation Partnership Project (3GPP) refers to a consortium of a number of standards organizations which develop protocols for mobile telecommunications. Its best-known work is the development and maintenance of -
 - ✓ GSM and related 2G standards, including GPRS and EDGE.
 - ✓ 3G standards, including HSPA and HSPA+
 - ✓ LTE and related 4G standards, including LTE Advanced and LTE Advanced Pro
 - ✓ 5G and related 5G standards.



1 MILLIMETRE WAVE SPECTRUM

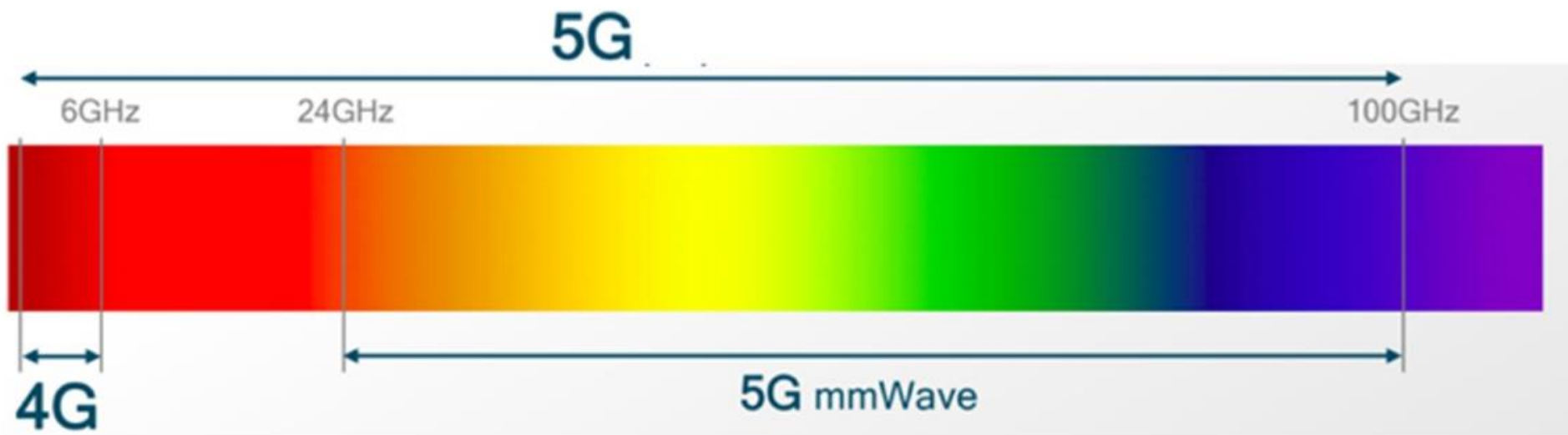
- The 5G networks will operate in the millimetre wave spectrum (30-300 GHz) which has the advantage of sending large amounts of data at very high speeds because the frequency is so high, it also experiences little interference from surrounding signals.
- The spectrum for 5G services not only covers bands below 6 GHz, including bands currently used for 4G LTE networks, but also extends into much higher frequency bands not previously considered for mobile communications.
- Although the mm Wave bands extend all the way up 300 GHz, it is the bands from 24 GHz up to 100 GHz that are used for 5G. And it is the use of this frequency bands that provide benefits for 5G networks.



FEATURES OF 5G TECHNOLOGY

1 MILLIMETRE WAVE SPECTRUM

- The high-frequency bands in the spectrum above 24 GHz have the potential to support large bandwidths and high data rates for increasing the capacity of wireless networks.
- So, 5G will bring wider bandwidths by expanding the usage of spectrum resources, from sub-3 GHz used in 4G to 100 GHz and beyond.
- 5G can operate in both lower bands (e.g., sub-6 GHz) as well as mmWave (24 GHz and up), which will bring extreme capacity, high speed and low latency.



FEATURES OF 5G TECHNOLOGY

2 UPGRADED LTE

- 5G is the latest upgrade in the long-term evolution (LTE) mobile networks.
- ✓ LTE (Long-Term Evolution) is a fourth-generation (4G) wireless standard that provides increased network capacity and speed for cell phones and other cellular devices compared with third-generation (3G) technology.
- ✓ LTE offers higher peak data transfer rates than 3G, initially up to 100 Mbps downstream and 30 Mbps upstream.



Speeds up to 1Gbps

Great for mobile
internet and HD video
streaming

Available on all
smartphones



Speeds up to 20Gbps

Designed for connected
homes and cities, AI, VR
and more

Only available on
selected mobile devices

FEATURES OF 5G TECHNOLOGY

3 INTERNET SPEED

- In the high-band spectrum of 5G, internet speeds have been tested to be as high as 20 Gbps (gigabits per second) as compared to the maximum internet data speed in 4G recorded at 1 Gbps.
- 5G network speeds should have a peak data rate of 20 Gb/s for the downlink and 10 Gb/s for the uplink.





DIFFERENCES BETWEEN 4G AND 5G

1 LATENCY

The biggest difference between the two is latency, which is simply the delay between commands and responses. 5G promises latency under 1 millisecond, while 4G latency varies from 60 to 95 milliseconds.

2 DOWNLOAD SPEEDS

With lower latency comes faster download speeds. The max download speed in 4G is 1 Gbps, while 5G can provide up to 10Gbps.

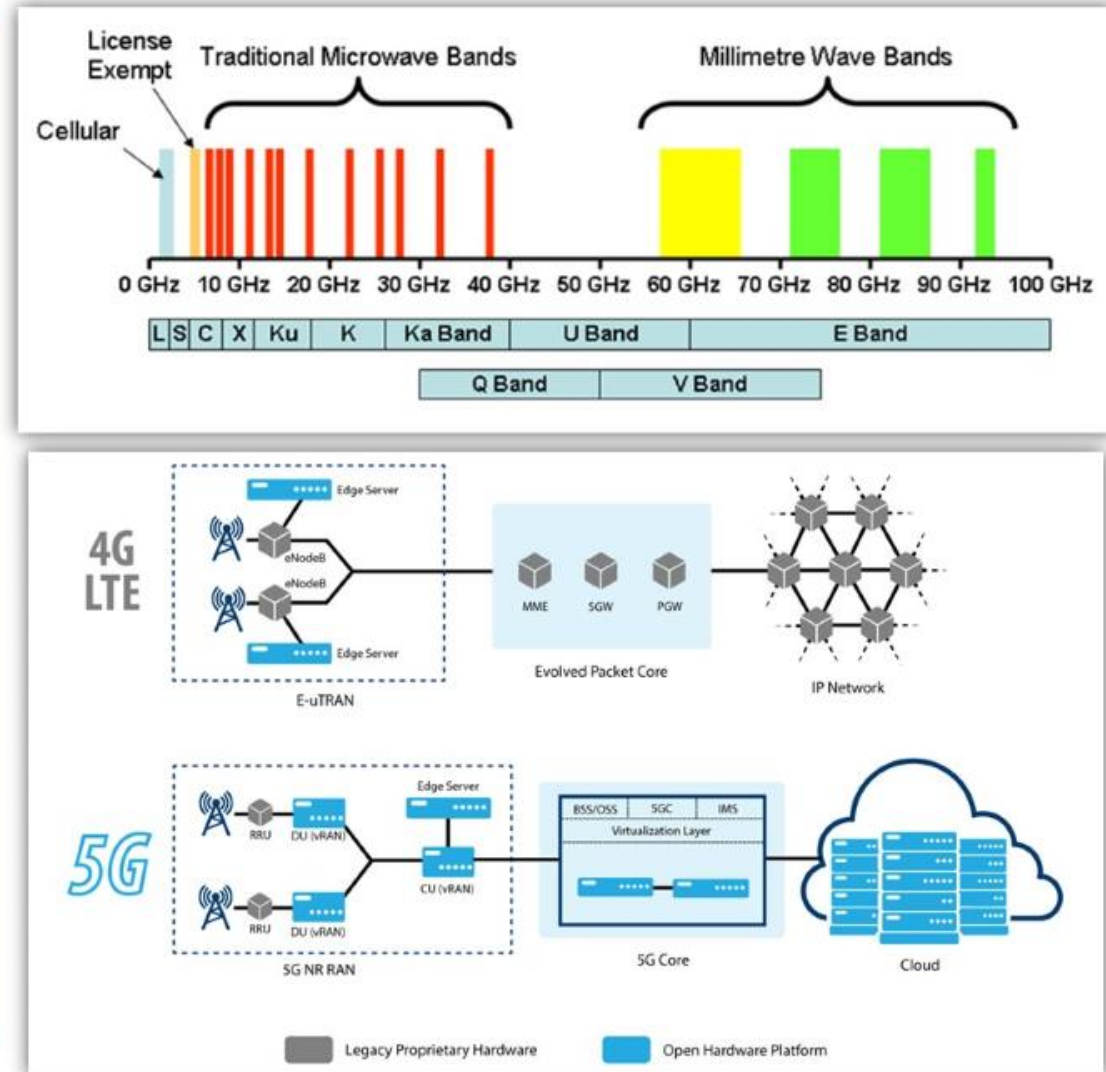
3 BASE STATIONS

- 4G like its predecessors transmits signals from large cell towers. 5G uses small cell technology to deliver the speeds it promises. Small cells, the size of a small box will deliver high frequencies but will have a smaller range. So, they must be deployed more densely than cell towers. So, higher Cell Density.

Because 5G networks will be denser, they can support more users and more types of devices.

DEPLOYMENT OF 5G IN INDIA

- Its deployment will happen with the auction of the millimetre-wave spectrum (the frequency range of 28 - 100 GHz). In this area India is not fully prepared and that's why 5G implementation will take more time.
- The switch from 4G to 5G will be infrastructure intensive i.e., mobile service providers will have to upgrade their underlying hardware countrywide in order to provide 5G speeds.
- Overnight upgradation of hardware is not possible so telecom companies would use the 4G LTE infrastructure to eventually migrate to 5G. It means that even when 5G is deployed, it will take the telecom giants, a couple of years to achieve the speed of 5G.



DEPLOYMENT OF 5G IN INDIA

- The customers will have to upgrade their SIM cards and buy new 5G enabled phones to access better speeds.
- India being a traditional late adopter of advanced technologies, is still struggling to achieve the speed which 4G network offers.
- In that scenario any major developments in coming year are doubtful. Though the Telecom Regulatory Authority of India (TRAI) has started the process of auctioning the 5G spectrum, analysts have predicted that India will only fully adopt 5G by 2022.



Telecom Regulatory
Authority of India

ADVANTAGES

- 5G is expected to offer enhanced mobile broadband that can meet high coverage requirements.
- It will offer very high upload and download speed.
- High data speed of 5G network might help cloud systems to stream software updates, music, and navigation data.
- 5G will help to incorporate Artificial Intelligence (AI) in our daily lives.
- It will also facilitate the ecosystem for the Internet of Things (IoT).
- 5G will help a lot in utilizing critical services like Telemedicine.
- It will help in digital growth of country that will result in rise of GDP and employment generation in the country.

For India 5G network could improve the accessibility of services such as mobile banking and healthcare, and enable exponential growth in opportunities for unemployed or underemployed people to engage in fulfilling and productive work. For this Government has rolled out enabling policies.



INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

GLOBAL PROGRESS ON 5G

- Global telecom companies have already started building 5G networks and rolling it out to their customers in many countries:
5G had been deployed in 50 cities in the United States.
- South Korea has rolled out 5G to 85 cities.
- Japan and China have too started 5G mobile service on a trial basis.



SOUTH KOREA

85



CHINA

57



USA

50

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

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98499 42299